

ALERT meeting

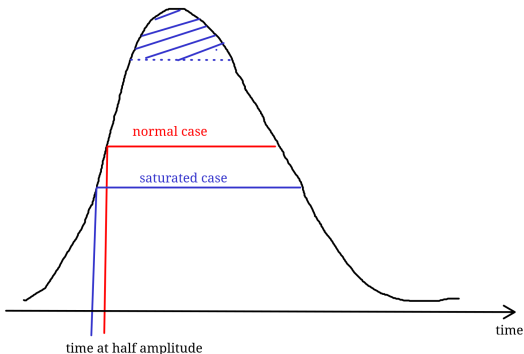
Time walk study

Felix Touchte Codjo

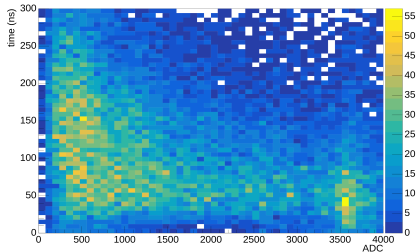
IJCLab

August 3, 2026

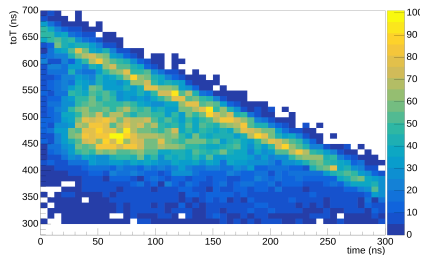
- ▶ Past weeks, I was looking at 4He DVCS in pass 0 data
- ▶ Remaining issues in the reconstruction has certainly reduced the number of events I was able to see
- ▶ Particularly, most 4He signals are associated to saturated AHDC pulses
- ▶ For such signals, the arrival time and the timeOverThreshold can be wrong



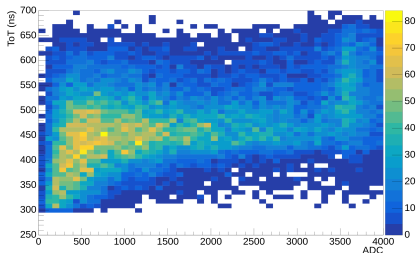
ADC vs time



time vs TimeOverthreshold



ADC vs timeOverThreshold



- ▶ We are particularly interested in early and saturated signals
- ▶ The plateau in time vs ToT are normal signals
- ▶ We also have a time walk effect on low ADCs

- ▶ measure ToT $\rightarrow$ correct ADC $\rightarrow$ deduce Time
- ▶ Several approaches
 - A fit (as illustrated, need to be improved)
 - Parametrization by comparing with the residual
 - Simulation
- ▶ Validation
 - Signals with ADC between 2500-3500

